

B.Sc. (Hons.) (Semester-V) Examination, 2020-21**Physics****Paper : BPT-505****Electromagnetic Theory****Time : Four Hours****Full Marks : 70****Instructions :**

1. The Question paper contains **08** questions out of which you are required to answer **any 04** questions. The question paper is of **70** marks with each question carrying **17.5** marks.
2. The total duration of the examination will be **4 hours** (Four hours), which includes the time for downloading the question paper from the Portal, writing the answers by hand and uploading the hand written answer sheets on the portal.
3. For the students with benchmark disability as per Persons with Disability Act, the total duration of examination shall be **6 hours** (six hours) to complete the examination process, which includes the time for downloading the question paper from the Portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.
4. Answer should be hand-written on a plain white **A4** size paper using **black** or **blue** pen. Each question can be answered in upto **350** words on **3** (Three) plain **A4** size paper (only one side is to be used).
5. Answer to each question should start from a fresh page. All pages are required to be numbered. You should write your Course Name, Semester, Examination Roll Number, Paper Code, Paper title, Date and Time of Examination of the first sheet used for answers.

Questions :

- Q.1.** In vacuum, let the scalar potential $V = 0$ and the vector potential $\vec{A} = A_0 \sin(kx - \omega t)\hat{a}_y$, where A_0 , ω and k are constants.
- (i) Calculate the electric field $\vec{E}$ and magnetic field $\vec{B}$ and check whether they satisfy Maxwell equations. **12.5**
 - (ii) If so, what conditions can be imposed on ω and k . **5**
- Q.2.** (i) Obtain an expression for the charge as well as the current densities in dielectric medium. **7.5**
- (ii) Using (i), construct the Maxwell equations and comment on the medium parameters. **10**
- Q.3.** 1. Derive the general expressions of the attenuation (α) and phase constants (β) for the conducting media. **5.5**
2. The magnetic field intensity of a linearly polarized uniform plane wave propagating in the +ve y -direction in the sea water [$\epsilon_r = 80.0, \mu_r = 1, \sigma = 4$ (S/m)] is $\vec{H} =$

$\hat{a}_x 0.1 \sin(10^{10}\pi t - \pi/3)$ (A/m) at $y = 0$. Determine the attenuation constant, the phase constant, the intrinsic impedance, phase velocity, wavelength and the skin depth. **12**

Q.4. 1. Given that $\vec{A} = 10 \cos(10^8 t - 10z - 60^\circ) \hat{a}_x$ and $\vec{B}_s = (20/j) \hat{a}_y + 10 e^{j2\pi/3} \hat{a}_y$. Express $\vec{A}$ in phasor form and $\vec{B}$ in instantaneous form. **7.5**

2. Show that in the case of perpendicular polarization (S-wave) of plane electromagnetic wave, $\sin^2 \theta_{B\perp} = \dots$ for non-magnetic media ($\mu_1 = \mu_2 = \mu_0$) where $\theta_{B\perp}$ the Brewster angle. **10**

Q.5. 1. Prove that for transverse electric mode in rectangular waveguide, the expression for z -component of magnetic field is given by $H_{zs} = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$, where symbols have their usual meaning. **8**

2. The phase velocity in the rectangular waveguide is greater than the velocity of light in the vacuum. Does this violate Einstein's relativity theory that messages cannot travel faster than the speed of light? **5.5**

3. A manufacturer produces a ferrite material with $\mu = 750\mu_0$, $\varepsilon = 5\varepsilon_0$, $\sigma = 10^{-2}$ S/m at 10 MHz. Would you classify the material as lossless, lossy or conducting? **4**

Q.6. 1. Show that product of intrinsic wave impedances for TE & TM modes of rectangular waveguide is μ/ε , where μ and ε are permeability and permittivity of lossless dielectric filled in the waveguide. **8.5**

2. A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1$ cm is to operate below 15.1 GHz. The waveguide is filled with a medium characterized by $\sigma = 0$, $\mu = \mu_0$, $\varepsilon_r = 1$. Calculate the phase constant, phase velocity and wave impedance for TE₁₀ mode at the operating frequency of 15 GHz. **9**

Q.7. 1. Is the field $\vec{E} = y^2 \hat{a}_x + (2xy + z^2) \hat{a}_y + (2yz) \hat{a}_z$ an electrostatic field? If so, obtain the potential. **7.5**

2. If a vector field $\vec{V} = y \hat{a}_x + (yz^2) \hat{a}_y + (3y + z) \hat{a}_z$, prove Stokes' theorem for a rectangular surface on the yz -plane with $y \rightarrow 0$ to 1 and $z \rightarrow 0$ to 2. **10**

Q.8. 1. If a spherical shell of radius R is uniformly charged, calculate its potential as well as the electrostatic field inside and outside the shell. **9**

2. If the magnetic field $\vec{B}$ is uniform, show that $\vec{A} = -\frac{1}{2}(\vec{r} \times \vec{B})$, where $\vec{r}$ is a vector from the origin to the point in the question. Is $\vec{B} = \vec{\nabla} \times \vec{A}$ and $\vec{\nabla} \cdot \vec{A} = 0$? Justify. **8.5**

Parameters to be used in the calculations :

Given $\mu_0 = 4\pi \times 10^{-7}$ H/m, $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

Charge of electron, $e = 1.6 \times 10^{-19}$ Coulomb

- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app>